Question - 6: Write a Python program to solve the Vacuum Cleaner problem.

Program Code

```
def vacuum_cleaner():
    rooms = {"A": "Dirty", "B": "Dirty"}
    location = "A"

    while "Dirty" in rooms.values():
        print("Vacuum is in Room", location)

        if rooms[location] == "Dirty":
            rooms[location] = "Clean"
            print("Room", location, "cleaned")

        if location == "A":
            location = "B"
        else:
            location = "A"

    print("All rooms are clean!")
    print(rooms)


vacuum_cleaner()
```

Program Output

```
PS E:\Desktop\SIMATS\Artificial In
ython/Python314/python.exe "e:/Des
m_cleaner.py"
Vacuum is in Room A
Room A cleaned
Vacuum is in Room B
Room B cleaned
All rooms are clean!
{'A': 'Clean', 'B': 'Clean'}
PS E:\Desktop\SIMATS\Artificial In
```